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1 * NOTE: idt() = integration function
2 .SUBCKT Block1_fixe_Drehzahl U1 V1 W1 F1 M n GND1
3 + PARAMS:
4 + UN_verkettet=20000
5 + IN=14.434k
6 + Pel=400e6 ;UN_verkettet*IN*sqrt(3)*cosphi of 0.8
7 + fn=50
8 + Kc=0.5
9 + UE0=600V ; Power of exciter = 0.25-5% * SN machine (smaller values for higher SN)
10 + IE0=2000A
11 + p=2
12 + Streuung=0.05 ;
13 + rcu=0.001
14 + rbg=0.001
15 + Schenkelpolfaktor=1 ;(1=cylindrical pole 0.5=Salient pole)
16 + Daempferfaktor=1
17 + err0=0
18 + RE_factor=0.6666
19 + n0=1500
20 + n1={n0} ;starts with 1500U/min
21 + alpha_0=0
22 + a=0.9 ; Parameter for nonlinear noload line (should be about 0.6-0.9)
23 + m=7
24 .PARAM
25 + a1={a}; Saturation parameter d-axis
26 + m1={m}; Saturation parameter d-axis
27 + a2={a1}; Saturation parameter q-axis
28 + m2={m1}; Saturation parameter q-axis
29 + UN={UN_verkettet/sqrt(3)}
30 + ZN={UN/IN}
31 + Rs={ZN*rcu}
32 + omega_el={(2*pi)*fn}
33 + Ls={Streuung*ZN/Kc/omega_el}
34 + PsiN={sqrt(2)*UN/omega_el}
35 + RE={UE0/IE0}
36 + RD={Rs/Daempferfaktor}
37 + RDx={RD} ; can be overwritten by setting this parameters on modelline
38 + RDy={RD} ; can be overwritten by setting this parameters on modelline
39 + LDx={Ls} ; can be overwritten by setting this parameters on modelline
40 + LDy={Ls} ; can be overwritten by setting this parameters on modelline
41 + LE={Ls} ; can be overwritten by setting this parameters on modelline
42 + R0={Rs} ; can be overwritten by setting this parameters on modelline
43 + L0={Ls} ; can be overwritten by setting this parameters on modelline (X0 is 0.03-0.1 p.u)
44 + Lsd={Ls}
45 + Lsq={Ls}
46 + XD={UN/IN/Kc}
47 + XQ={XD*Schenkelpolfaktor}
48 + g={UE0/(UN*sqrt(2)*RE/(XD-omega_el*Lsd))} ; updated formula for g parameter
49 + g1={UE0/(UN*sqrt(2)*RE/(XQ-omega_el*Lsq))} ; updated formula for salient pole machine
50 + ImNx={IE0/g} ; d-axis magnetizing current
51 + ImNy={IE0/g1} ;q-axis magnetizing current
52 + MN={Pel*(1-rbg-rcu)*p/omega_el}
53 *
54 *      currents of windings
55 G1  U1  U2  VALUE={v(isxS)+v(i_Null)}
56 G2  V1  V2  VALUE={{-v(isxS)+v(isyS)*sqrt(3))/2+v(i_Null)}
57 G3  W1  W2  VALUE={{-v(isxS)-v(isyS)*sqrt(3))/2+v(i_Null)}

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58 GiE      F1      0      VALUE={v(ie)*g}
59 *          voltage components
60 EuxS  usxS  0  VALUE={{(2*v(u1,u2)-v(v1,v2)-v(w1,w2))/3}}
61 EuyS  usyS  0  VALUE={{(v(v1,v2)-v(w1,w2))/sqrt(3)}}
62 Eux    usx    0  VALUE={ v(usxS)*cos(v(alpha))+v(usyS)*sin(v(alpha))}
63 Euy    usy    0  VALUE={-v(usxS)*sin(v(alpha))+v(usyS)*cos(v(alpha))}
64 * Starpoint
65 R2    v2  GND1  1u
66 R12   u2  GND1  1u
67 R23   w2  GND1  1u
68 RGND1 GND1 0 1e6
69 *          Zerosystem
70 Eu0   u_Null 0  VALUE={{(v(u1,u2)+v(v1,v2)+v(w1,w2))/3}}
71 Ei0   i_Null 0  VALUE={idt(v(u_Null)-R0*v(i_null))/LO}
72 *          current components
73 EisxS  isxS  0  VALUE={v(isx)*cos(v(alpha))-v(isy)*sin(v(alpha))}
74 EisyS  isyS  0  VALUE={v(isy)*cos(v(alpha))+v(isx)*sin(v(alpha))}
75 *          magnetic coupling of direct axis with saturation
76 Gpsix  0  isx  VALUE={v(psisx)}
77 Rsx    isx  0  {1/Lsd}
78 Gpsidx isx iDx1 VALUE={v(psiDx)} ;
79 Rsdx   iDx1 isx {1/LDx}
80 GpsiEx iDx1 imx1 VALUE={v(psiE)}
81 RsEx   imx1 iDx1 {1/LE}
82 vphx   imx1 imxa1 0
83 Eimx   imxa1 0  VALUE={ImNx*(a1*i(vphx)/PsiN+(1-a1)*PWRS(i(vphx)/PsiN,m1))}
84 *          magnetic coupling of quadrature axis with saturation
85 Gpsiy  0  isy  VALUE={v(psisy)}
86 Rsy    isy  0  {1/Lsq}
87 Gpsidy isy imy1  VALUE={v(psiDy)}
88 Rsdy   imy1 isy {1/LDy}
89 vphy   imy1 imya1 0
90 Eimy   imya1 0  VALUE={ImNy*(a2*i(vphy)/PsiN+(1-a2)*PWRS(i(vphy)/PsiN,m2))}
91 *          damper currents
92 Eidx  idx 0  VALUE={v(idx1)-v(isx)}
93 Eidy  idy 0  VALUE={v(imy1)-v(isy)}
94 *          exciter current
95 Eie  ie 0  VALUE={v(imx1)-v(idx1)}
96 *          Flux components
97 Epsi_sx psisx 0  VALUE={err0*PsiN+idt(v(usx)-Rs*v(isx)+p*(2*pi)/60*v(n)*v(psisy))}
98 Epsi_sy psisy 0  VALUE={idt(v(usy)-Rs*v(isy)-p*(2*pi)/60*v(n)*v(psisx))}
99 Epsi_Dx psiDx 0  VALUE={err0*PsiN+idt(-RDx*v(idx))}
100 Epsi_Dy psiDy 0  VALUE={idt(-RDy*v(idy))}
101 Epsi_E  psiE 0  VALUE={err0*PsiN*(1+Streuung)+idt(v(F1,0)*g*RE_factor-RE*RE_factor*g*g*v(ie))}
102 *          Load angle
103 Etheta 0 theta VALUE={idt(180/pi*atan2(v(usx),v(usy)))}
104 *          inner torque
105 Emi  Mi 0  VALUE={1.5*p*(i(vphx)*v(isy)-i(vphy)*v(isx))}
106 *          outhertorque
107 EMoment M 0  VALUE={-v(Mi)+v(Rbg)}
108 Vn  n 0 {n1}
109 * friction
110 GRbg 0 Rbg TABLE {v(n)} (-1,-1) (1,1)
111 RRbg  Rbg 0 {MN*rbg}
112 * Compute angle alpha
113 Ealpha alpha 0  VALUE={alpha_0*pi/180+idt(v(n))*p/60*(2*pi)}
114 .ENDS

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